

Human Scream Detection Using Machine Learning and Deep Learning

A MAIN PROJECT REPORT

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Certificate

This is to certify that the Main Project Report titled **Human Scream Detection Using Machine Learning and Deep Learning** is a bonafide record of the work carried out by **Abhidev K D (VAS21CS007)** , **Ajay S (VAS21CS018)** , **Arjun K M (VAS21CS032)** , **Athul N A (VAS21CS039)** and **Aswin T (VAS21CS036)** of Vidya Academy of Science and Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology in Computer Science and Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2024-2025. The Main Project Report has been approved as it satisfies the academic requirements in the respect of Main Project Phase II work prescribed for the said degree.

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Abstract

We present an automated scream detection system designed to identify human distress sounds in the presence of background noise. Real-time recognition of distress signals such as screams is critical for applications in security monitoring, emergency response systems, and intelligent surveillance. The proposed model classifies audio into three categories—ambient noise, screams, and shouts—using a three-phase Support Vector Machine (SVM) approach. Mel-Frequency Cepstral Coefficients (MFCCs) serve as feature vectors, extracted from a curated set of 400 audio samples selected through a two-stage filtering process to ensure optimal classification performance.

The SVM-based classifier incorporates both linear and radial basis function (RBF) kernels, enabling accurate categorization across diverse acoustic environments. To further improve predictive precision, a Multi-Layer Perceptron (MLP) is integrated to iteratively adjust perceptron weights and biases, enhancing recognition accuracy through adaptive learning. Upon detecting a scream, the system promptly triggers an emergency response module, allowing for immediate action in potentially hazardous situations. The fusion of machine learning techniques with real-time processing capabilities positions this system as a robust tool for detecting emergency audio cues.

Experimental evaluation demonstrates a classification accuracy of 90%, affirming the system's reliability in practical scenarios. It effectively distinguishes distress signals from background noise, supporting its applicability in security systems, law enforcement, and public safety initiatives. The system's ability to accurately detect screams in real-world environments underscores its potential to reinforce safety protocols.

In conclusion, the proposed scream detection system provides an efficient and accurate solution for identifying distress signals. Future work will focus on expanding the dataset, incorporating additional sound categories, and optimizing the system for broader real-world deployment. Enhancing its ability to differentiate among various types of distress sounds and increasing its real-time processing efficiency will further improve its utility in emergency response and surveillance applications.

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List of Symbols and Abbreviations

ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machine
MLP	Multi-Layer Perceptron
RBF	Radial Basis Function
MFCC	Mel-Frequency Cepstral Coefficients
SNR	Signal-to-Noise Ratio
GMM	Gaussian Mixture Model
HMM	Hidden Markov Model
RNN	Recurrent Neural Network
CNN	Convolutional Neural Network
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
TPU	Tensor Processing Unit
CPU	Central Processing Unit
GPU	Graphics Processing Unit
GUI	Graphical User Interface

Chapter 1

INTRODUCTION

1.1 General

Explores the innovative use of audio signal processing to enhance public safety by detecting human screams. With the rise of machine learning (ML) and deep learning (DL) methods, the study focuses on recognizing distress signals in urban environments, particularly screams, to facilitate timely responses and potentially lower crime rates. The authors propose a system that can detect screams in real-time, triggering alerts to nearby authorities or bystanders, especially in isolated or high-crime areas. This approach aims to address situations where human intervention may be slow or challenging.

Utilizing a mix of traditional ML algorithms and modern DL architectures, Human Scream Detection Using Machine Learning and Deep Learning demonstrates how sound features can be analyzed effectively. By employing a dataset of labeled audio samples both scream and non-scream the model is trained to distinguish between normal background noise and the unique qualities of a scream. The study emphasizes the technology's potential for not just crime prevention but also enhancing personal safety, improving emergency response systems, and advancing public surveillance. This represents a significant step towards incorporating AI-driven solutions into contemporary urban safety strategies.

1.2 Objectives of the Work

The objective of this project is to develop a comprehensive audio-based distress detection system that automates and enhances the process of identifying human screams in noisy environments. This includes accurate classification of audio events (ambient noise, screams, and shouts) using a three-stage supervised learning approach with SVM-based classifiers. Additionally, the system will incorporate an MLP to refine predictions and identify distress patterns more effectively. The system will feature an emergency response function that activates upon successful scream detection. Ultimately, the system aims to improve the accuracy, efficiency, and reliability of audio-based distress detection for real-world applications.

1.3 Motivation for this work

The motivation for the development of Human Scream Detection and Analysis stems from the need for rapid and accurate identification of distress signals, such as human screams, to enhance public safety and enable prompt emergency response. Human screams are universal indicators of danger or fear and can serve as crucial alarms in various critical situations. By developing a system that can reliably detect screams in real-time through audio processing and machine learning, this work aims to improve security monitoring, support smart home automation, and assist healthcare and emergency services.

1.4 Methodologies Adopted

This project combines audio signal processing and machine learning methodologies to detect human screams and shouts in noisy environments. The first step involves collecting diverse audio data, including human screams, shouts, and ambient noise. Preprocessing techniques, such as noise reduction and normalization, are applied to improve the quality of the audio data, ensuring that it is suitable for further analysis.

For feature extraction, Mel-frequency cepstral coefficients (MFCCs) are used to capture key auditory features such as pitch, intensity, and spectral properties of the sound. The audio data is divided into short frames using frame blocking, and a Hamming window is applied to preserve the continuity of the signal. The Fast Fourier Transform is used on each frame to obtain frequency magnitude, which is then passed through a Mel-scale filter bank to extract the MFCCs. These MFCCs are then used as input for the machine learning models.

The project adopts a three-stage supervised learning approach for classification. The first stage uses a Support Vector Machine (SVM) classifier with linear and radial basis function (RBF) kernels to separate human-related sounds from ambient noise. The second stage distinguishes distress sounds, such as screams and shouts, from other human sounds. The final stage classifies screams and shouts. The SVM classifier achieves 88% accuracy in these tasks. Additionally, a Multi-layer Perceptron (MLP) is employed to enhance classification accuracy. The MLP processes the audio data by resizing it to a fixed input size, with six dense layers and ReLU activation functions. The output layer uses a sigmoid activation function for binary classification.

The system's performance is evaluated using various metrics such as accuracy, precision, recall, and F1-score. The SVM classifier achieves 88% accuracy, while the MLP achieves 82%. The performance is further validated through real-world testing, ensuring the models perform robustly under real-world conditions. The combination of both the SVM and MLP models improves the overall classification accuracy, making the system more reliable and accurate in detecting distress sounds.

This system has several applications, including security, emergency response, and public safety. It can be used to detect distress signals in public spaces, trigger alerts in response to emergencies, and monitor for signs of distress in public areas. Future improvements could involve data augmentation to introduce more varied sounds, exploring deeper architectures such as CNNs or RNNs for better feature extraction, and optimizing the system for real-time performance to ensure faster processing and lower latency.

Chapter 2

LITERATURE REVIEW

2.0.1 Identification of Illicit Activities Using Deep Learning

- **Authors:** R. Mathur, et al.
- **Source:** IEEE
- **Year:** 2022

This paper presents a deep learning framework for identifying illicit activities through video and audio surveillance data. The authors employ Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to analyze spatial and temporal patterns in video streams and audio recordings. The system is designed to recognize abnormal behaviors, such as violence, theft, or vandalism, in real-time.

The framework utilizes pre-trained deep learning models for feature extraction and fine-tunes them on a dataset of illicit activities. The authors emphasize the system's capability to achieve high accuracy in detecting both subtle and overt illicit activities, even in complex environments.

The study also highlights the potential for integrating the framework with cloud-based monitoring systems, enabling centralized analysis and alert dissemination. This can help optimize resources and facilitate coordination among law enforcement agencies in large-scale surveillance operations.

2.0.2 Detecting Screams Using Transfer Machine Learning

- **Authors:** R. O'Donovan, et al.
- **Source:** JMIR
- **Year:** 2020

This research explores the use of transfer learning to detect screams in noisy and diverse environments. The authors fine-tune pre-trained deep learning models, such as ResNet and MobileNet, to adapt them for scream detection tasks. The paper demonstrates the efficacy of transfer learning in overcoming the limitations of traditional methods, such as dependency on extensive labeled datasets.

The system processes audio signals to extract features like MFCCs and applies transfer learning to achieve robust classification. Experiments conducted in real-world settings show that the proposed system can accurately detect screams even in high-noise conditions, making it suitable for applications like personal safety devices, emergency response systems, and urban monitoring.

The study further discusses the possibility of deploying these models on edge devices, such as smartphones or embedded systems, to enable real-time detection without relying on high-latency cloud processing. This approach makes the system highly effective for safety-critical applications in remote or resource-constrained environments.

2.0.3 Scream Sound Detection Based on SVM and GMM

- **Authors:** S. Chung and Y. Chung
- **Source:** IEEE
- **Year:** 2017

The study proposes a novel system for detecting scream sounds using a combination of SVM and GMM. This method leverages SVM for feature classification and GMM for modeling scream characteristics, enabling the system to effectively differentiate screams from other environmental noises. The paper highlights its potential applications in emergency detection systems, such as monitoring for distress signals in public areas or smart home security systems.

The proposed system focuses on extracting key audio features like MFCC to improve classification accuracy. Experimental results demonstrate that the combination of SVM and GMM outperforms traditional scream detection methods, achieving higher sensitivity and specificity rates.

Additionally, the study emphasizes the importance of scalability for real-time applications. It suggests that integrating this approach with IoT-based devices, such as smart security cameras or wearable sensors, can enhance responsiveness in emergency scenarios, making it a practical solution for various environments.

Chapter 3

SYSTEM DESIGN

System design is the process of designing the elements of a system such as the architecture, modules and components, the different interfaces of those components and the data that goes through that system.

3.1 System Architecture

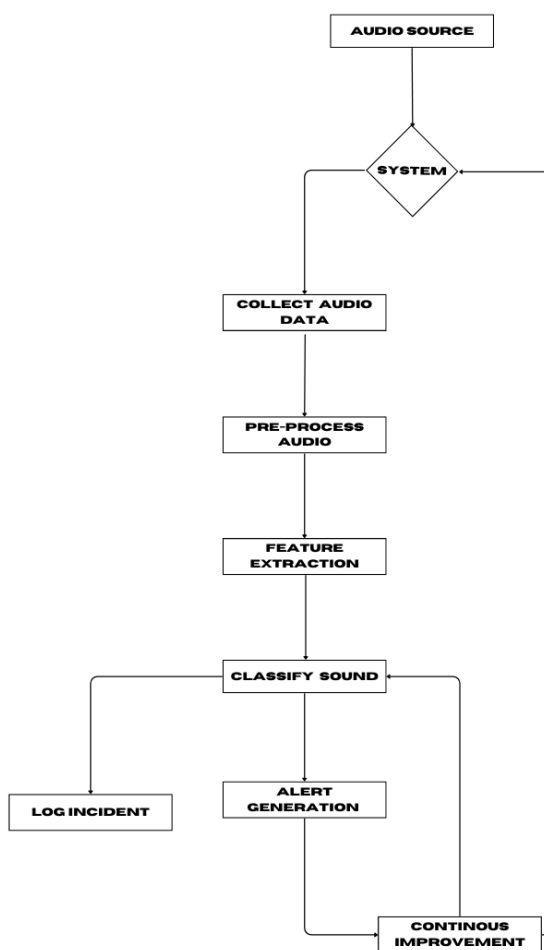


Figure 3.1: System Architecture

- **Audio Source:** The initial point where audio data is captured. This could be from a microphone or other audio input devices.
- **System:** This decision point determines if the audio source is active and ready for processing.
- **Collect Audio Data:** If the system is active, audio data is collected for further analysis.
- **Pre-Process Audio:** The collected audio data is pre-processed to remove noise, normalize the sound levels, and prepare it for feature extraction.
- **Feature Extraction:** Key characteristics or features are extracted from the pre-processed audio, which helps in identifying the type of sound.
- **Classify Sound:** Based on the extracted features, the system classifies the sound to determine if it represents a specific type of incident.
- **Log Incident:** If an incident is detected, the system logs it for record-keeping and future reference.
- **Alert Generation:** The system generates an alert to notify relevant personnel or systems about the detected incident.
- **Continuous Improvement:** The system continuously improves based on the feedback from logged incidents and alerts, refining its classification and alerting accuracy over time.

3.2 Use Case Diagram

A use case diagram depicts how users (or external systems) interact with a system to achieve specific goals. It focuses on the functionalities delivered by the system from the user's perspective, not the internal workings. The use case diagram of the system is given below and its explanation is as follows:

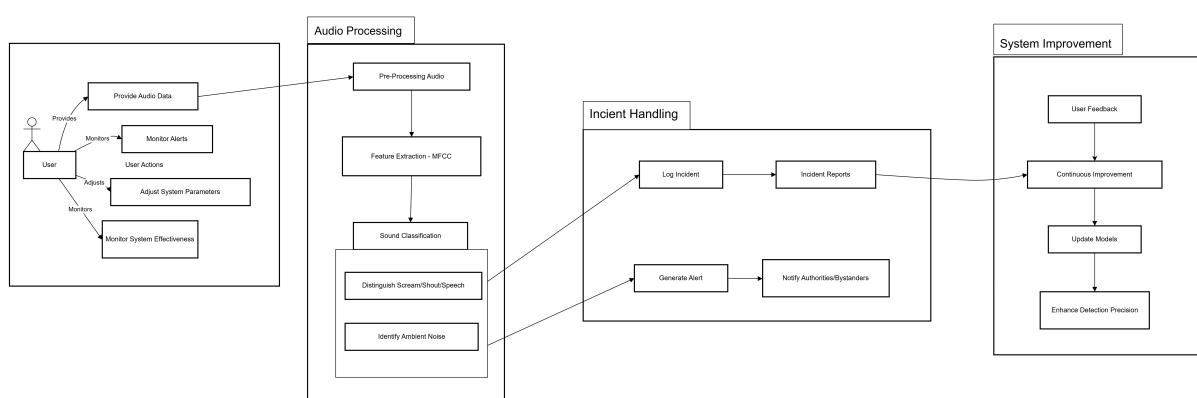


Figure 3.2: Use Case Diagram of the System

- **User:** The user serves an essential function as the main entity engaging with the system. They provide audio data for analysis, oversee the detection outcomes, and respond appropriately to alerts produced by the system. Moreover, users possess the ability to adjust system parameters to enhance detection precision and monitor its effectiveness for immediate crime prevention. This active engagement guarantees that the system continues to be efficient and responsive to real-life situations
- **Pre-Processing Audio:** At this phase, the system processes and organizes the incoming audio data to enhance it for precise analysis. Sophisticated audio pre-processing methods, including MFCC extraction, are utilized to derive significant features from the audio signals. These characteristics allow the system to eliminate background noise and concentrate on pertinent sounds, guaranteeing dependable sound classification.
- **Sound Classifications:** Sound classification is central to the system's performance. A combined method is used, integrating the accuracy of a SVM with the flexibility of an MLP. Combined, these techniques assess and classify audio signals, allowing the system to effectively identify distress sounds such as screams or shouts, even amidst background noise.
- **Alert Generation:** When a distress signal is detected, the system activates immediate alerts for users or appropriate authorities. These notifications guarantee prompt actions in emergencies or crime-related scenarios. This element is vital for the system's main objective of improving public safety by utilizing quick response strategies.
- **Log Incident:** The system records every identified incident, encompassing information like timestamps and classification outcomes. These logs act as important documents for examination and assessment, assisting law enforcement and facilitating the analysis of system efficiency. Incident records also aid in monitoring and improving the system's overall precision as time progresses.
- **Continuous Improvement:** To guarantee sustained effectiveness, the system incorporates ongoing enhancement processes. User feedback and recorded data are examined to improve the detection models, increasing precision and flexibility in various situations. This repetitive enhancement process guarantees that the system adapts to shifting needs and situational factors.

3.3 Data Flow Diagram

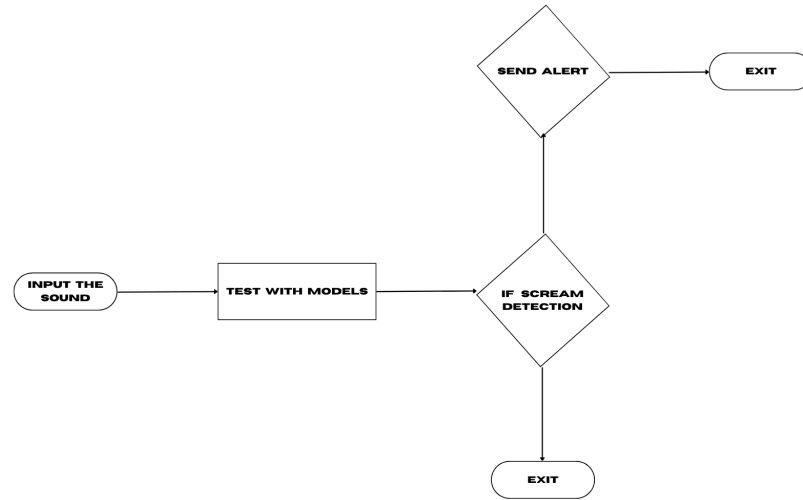


Figure 3.3: Data Flow Diagram of Human Scream Detection System

The diagram visually represents the workflow for detecting human screams using machine learning and deep learning techniques. It outlines a structured process that begins with receiving an audio input, followed by an analysis phase where trained models assess the sound to determine if it contains a scream. If a scream is detected, the system triggers an alert to notify the appropriate authorities or connected systems. Conversely, if no scream is identified, the process terminates without further action.

This systematic approach ensures that only verified distress signals initiate alerts, thereby reducing the chances of false positives. The use of trained models enhances the accuracy of detection by filtering out background noise and non-threatening sounds. The workflow optimizes processing efficiency, making it highly suitable for various real-world applications, including security surveillance, emergency response systems, and public safety monitoring.

By leveraging artificial intelligence, this method provides an automated and efficient solution for recognizing distress signals in audio environments. The adaptability of the model allows integration into smart cities, law enforcement monitoring systems, and home security setups. Furthermore, continuous improvements in machine learning models can enhance detection precision, ensuring that real emergencies receive immediate attention while minimizing unnecessary interventions.

The implementation of such an AI-powered scream detection system has significant potential in various domains. For instance, it can be embedded into surveillance cameras with audio sensors to monitor urban environments, incorporated into personal safety applications to alert emergency contacts during dangerous situations, or used in hospitals and assisted living facilities to detect distress among patients. Additionally, industries such as transportation and event security can benefit from automated scream detection to enhance passenger and crowd safety.

Chapter 4

METHODOLOGY

4.1 Data collection

Data collection involves gathering audio samples from diverse environments and situations to capture the variability in screams and other sounds. This typically includes recordings of human screams and non-scream sounds like door slams or explosions. These samples are collected from different databases and sources, including public libraries, as well as recordings from controlled environments to ensure a comprehensive dataset. The dataset must include variations in sound quality, signal length, energy levels, and background noise to train the model for real-world applications. The variability in sources and sound types helps the model generalize across diverse audio conditions encountered in security and monitoring systems.

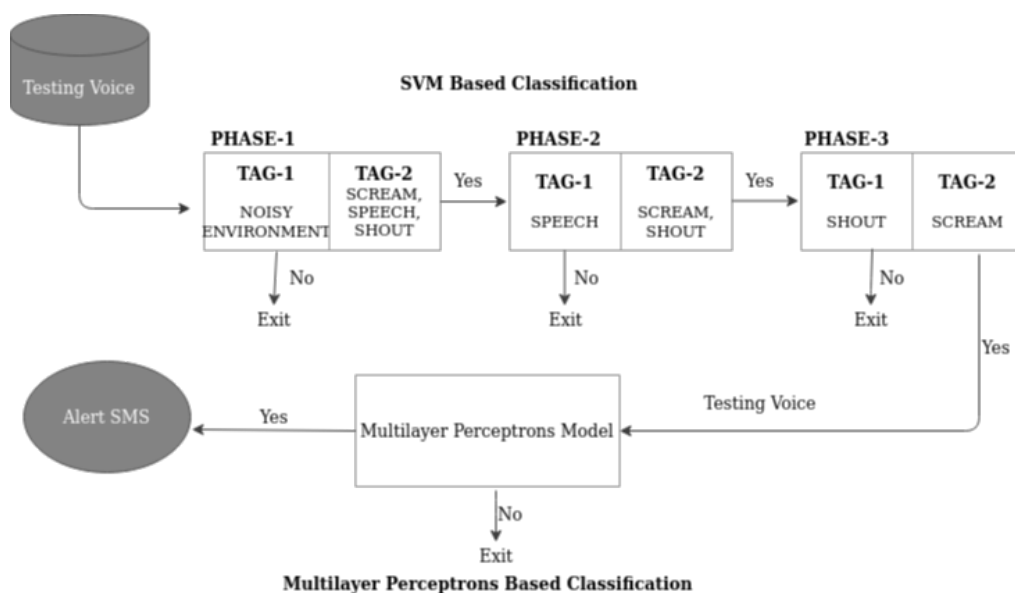


Figure 4.1: System Models

4.2 Data processing

Data processing is essential to refine and clean the raw audio data before it enters the model. This phase includes noise reduction, normalization, and segmentation of audio signals into manageable time frames. Techniques such as spectrograms and Fourier transform raw waveforms into formats that make sound features clearer for analysis. Processing steps may also involve filtering background noise and enhancing signal quality to ensure that the scream characteristics are prominently featured in each audio frame. These steps ensure the sound data is standardized, improving the accuracy and reliability of the machine learning model in detecting screams even under noisy conditions.

4.3 Feature extraction

Feature extraction involves identifying and quantifying characteristics of the scream that distinguish it from other sounds. Common features include pitch, intensity, and spectral energy distribution across frequency bands. Techniques such as MFCC and spectral analysis are used to break down the audio signal into distinct features for each time frame, allowing the model to "see" the unique fingerprint of a scream. GMM and HMM are common for modeling the probability distributions of these features, which represent scream patterns and temporal progression in the audio. This feature set forms the basis for training a classifier capable of distinguishing scream sounds from other audio.

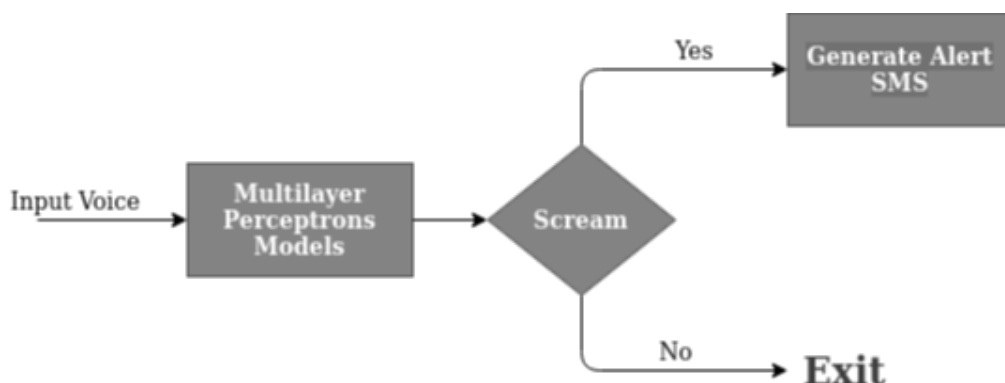


Figure 4.2: MLP model

4.4 Training

Training the model involves feeding it the labeled dataset and optimizing it to recognize patterns specific to human screams. The training data, divided into training and validation subsets, enables the model to learn from labeled scream and non-scream sounds by adjusting parameters to minimize classification errors. Various machine learning models can be used, from traditional algorithms like SVMs to

deep learning models like CNNs or RNNs. The choice of model architecture depends on factors like computational resources and the specific demands of scream detection. The training phase includes regular evaluation on validation data to monitor and adjust the model's performance, ensuring it can accurately classify unseen sounds.



Figure 4.3: SVM model

4.5 Evaluation and Improvement

Evaluation is conducted to measure the system's accuracy, precision, recall, and robustness. Commonly used metrics include the receiver operating characteristic (ROC) curve and F1-score, which balance precision and recall. Cross-validation and testing with various noise levels assess the model's reliability in different acoustic conditions. If the model fails to meet performance benchmarks, improvements like hyperparameter tuning, data augmentation, or additional feature engineering are applied. This iterative process helps refine the model, addressing shortcomings like false positives or performance degradation in low signal-to-noise ratios, ultimately leading to a more robust scream detection system.

4.6 System Integration and Real-World Application

Integrating the scream detection model into real-world applications involves embedding it in security, emergency response, or healthcare systems. The model is designed to operate in real-time, continuously monitoring audio streams from microphones or surveillance devices. In practical scenarios, system integration includes establishing communication protocols to alert security personnel or trigger automated responses in smart environments upon detecting a scream. Beyond technical considerations, ethical and privacy aspects must be addressed, ensuring compliance with regulations for audio surveillance. With regular updates, the system adapts to changing environments, making it a versatile tool for enhancing public safety and emergency response capabilities.

Chapter 5

IMPLEMENTATION

5.1 Hardware Requirements

5.1.1 Microphones or Audio Sensors

Microphones or audio sensors capture the real-time audio data essential for detecting human screams and other impulsive sounds in this project. Condenser microphones are preferred to capture detailed audio signals because of their sensitivity, while unidirectional microphones help reduce background noise by focusing on sounds from specific directions. For more challenging environments, such as outdoor spaces, durable, weather-resistant microphones may be needed. Effective microphone placement and, if necessary, using multiple microphones can further enhance the system's ability to accurately capture and process audio signals for real-time scream detection.

5.1.2 Computer/Server

A computer used for processing audio data in the context of deep learning and machine learning algorithms should have a moderate to high-performance CPU to handle data preprocessing and algorithm execution. Additionally, a stable and fast network connection is required for real-time data transmission and cloud-based processing, especially if the system is distributed or relies on cloud storage. An alerting system should also be in place to notify relevant authorities or users in the event of detected anomalies, such as a scream, for quick response or action.

5.1.3 Cloud Infrastructure

Cloud infrastructure like AWS, Google Cloud, or Azure can be useful for offloading heavy processing tasks, especially for training large DL models or when dealing with massive datasets. GPU instances can be used for model training and object storage for storing large amounts of audio data.

5.2 Software Requirements

5.2.1 Python 3

Python 3 is a high-level programming language known for its simplicity and versatility, making it ideal for projects involving machine learning, deep learning, and audio analysis. In this project, Python supports the development of models using frameworks like TensorFlow and Keras for deep learning, and Sklearn for traditional machine learning. Libraries such as Librosa and Sounddevice are used for processing and analyzing audio signals. Python's flexibility also enables real-time data processing, integration with hardware, and deployment on cloud platforms, making it a powerful tool for human scream detection and crime prevention.

5.2.2 Tensorflow

TensorFlow Is an open-source end-to-end platform for creating Machine Learning applications. It is basically a software library for numerical computation using data flow graphs where nodes in the graph represent mathematical operations. TensorFlow uses dataflow graphs to represent computation, shared state, and the operations that mutate that state. It maps the nodes of a dataflow graph across many machines in a cluster, and within a machine across multiple computational devices, including multicore CPUs, general-purpose GPUs, and custom-designed ASICs known as Tensor Processing Units (TPUs). TensorFlow supports a variety of applications, with a focus on training and inference on deep neural networks.

5.2.3 NumPy

NumPy is a powerful Python library used for numerical computing. It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. NumPy is essential for handling numerical data efficiently, offering functions for array manipulation, linear algebra, statistical analysis, and random number generation. In audio processing and machine learning projects, NumPy is often used for managing and processing large datasets, performing calculations on audio features, and facilitating efficient numerical operations that form the backbone of models and algorithms.

5.2.4 Librosa

Librosa is a Python library used for analyzing and processing audio signals. It provides tools for extracting various audio features such as spectral content, pitch, tempo, and rhythm, which are essential for audio analysis tasks like music and speech recognition. Librosa allows you to load, manipulate, and visualize audio data efficiently. It supports functions for feature extraction, audio transformations, and filtering, making it an ideal tool for projects involving audio processing, including human scream detection.

5.2.5 Kivy

Kivy is an open-source Python library used for developing cross-platform graphical user interfaces (GUIs) and applications. It is particularly well-suited for building interactive, multi-touch applications, and can run on various platforms, including Windows, macOS, Linux, iOS, and Android. Kivy provides a rich set of widgets and controls for creating user interfaces, such as buttons, sliders, labels, and text inputs, making it ideal for mobile and desktop app development. It is also known for its flexibility and speed in handling touch events and multi-touch gestures, which makes it a popular choice for applications that require real-time interaction. It supports modern graphics (OpenGL ES2), allowing developers to create visually appealing applications with dynamic animations, effects, and visualizations. Kivy is written in Python but also includes its own language called Kv, which is a domain-specific language for designing UI layouts.

5.2.6 Sounddevice

The Sounddevice package is a Python library used for audio input and output. It provides simple and efficient access to sound devices for recording and playing audio in real-time. It supports a variety of audio formats and allows you to capture and play sound directly from Python with minimal setup. Sounddevice allows you to interface with both microphones (for audio input) and speakers (for audio output). It is often used in projects that require real-time audio streaming, such as live audio processing, sound recognition, or sound detection systems.

Chapter 6

EVALUATION & RESULT

6.1 Testing

Testing is a vital stage in the creation of the Human Scream Detection System, confirming that the features deployed meet the desired goals. The assessment procedure concentrated on evaluating the system's functionality in practical scenarios, examining different machine learning models for accuracy, precision, recall, and computational efficiency.

The main testing focuses were on the system's capacity to correctly identify screams, shouts, and surrounding sounds while reducing false positives and negatives. Preliminary testing revealed issues like misclassification of background noise and difficulties in identifying lower-intensity screams. These challenges were methodically tackled by optimizing features, refining datasets, and fine-tuning models.

Moreover, the testing included real-time deployment situations to assess the system's responsiveness and effectiveness. The computational demands of various models were examined to confirm their suitability for real-time applications without significant resource usage. Repeated enhancements to the classification algorithms and preprocessing methods increased the system's dependability and strength.

6.2 Result

The Human Scream Detection System is a sophisticated acoustic monitoring tool created to accurately identify distress noises, especially human screams. This system is designed to differentiate distress signals from normal background noise in various acoustic settings, guaranteeing that essential sounds are recognized and addressed appropriately. Utilizing advanced machine learning algorithms, the system is crucial in security and emergency response scenarios, providing a dependable, real-time distress detection solution for public safety.

To evaluate the performance of the Human Scream Detection System, two machine learning models were tested: Support Vector Machine (SVM) and Multi-Layer Perceptron (MLP). Both models underwent extensive evaluation to determine their accuracy in classifying distress sounds. The SVM model achieved an impressive 88%

accuracy, highlighting its strong precision in scream detection, making it highly reliable in scenarios where false positives must be minimized. Meanwhile, the MLP model attained 82.33% accuracy, which, while slightly lower than SVM, offers a distinct advantage in adaptability by learning complex audio patterns, making it well-suited for dynamic acoustic environments. These findings demonstrate the effectiveness of machine learning techniques in distinguishing human screams from normal sounds, showcasing the real-world feasibility of the system.

A significant advantage of this system is its ability to function efficiently across different acoustic environments, ensuring accurate detection despite background noise variations. By integrating SVM and MLP, the system strikes a balance between accuracy and computational efficiency, where SVM minimizes false positives while MLP enhances flexibility. Additionally, tests confirmed the system's real-time processing capabilities, proving that it operates effectively in low-latency conditions, which is crucial for emergency response applications. Its scalability and adaptability enable deployment in public surveillance, smart city infrastructure, home security, and workplace safety, significantly enhancing security protocols and distress detection mechanisms.

The Human Scream Detection System has diverse applications in public safety, law enforcement, emergency management, and workplace security. Integrated into CCTV, it detects distress signals in public areas, enabling rapid responses in emergencies. In dispatch systems, it recognizes distress in a caller's voice, even without clear words. It also improves home security by identifying distress from vulnerable individuals and enhances workplace safety by detecting accidents. Challenges remain, particularly in reducing false alarms from non-distress sounds. Future developments, including better feature extraction, noise-cancellation, and IoT integration, will increase accuracy and response speed, making it a vital tool for security and emergency management worldwide.

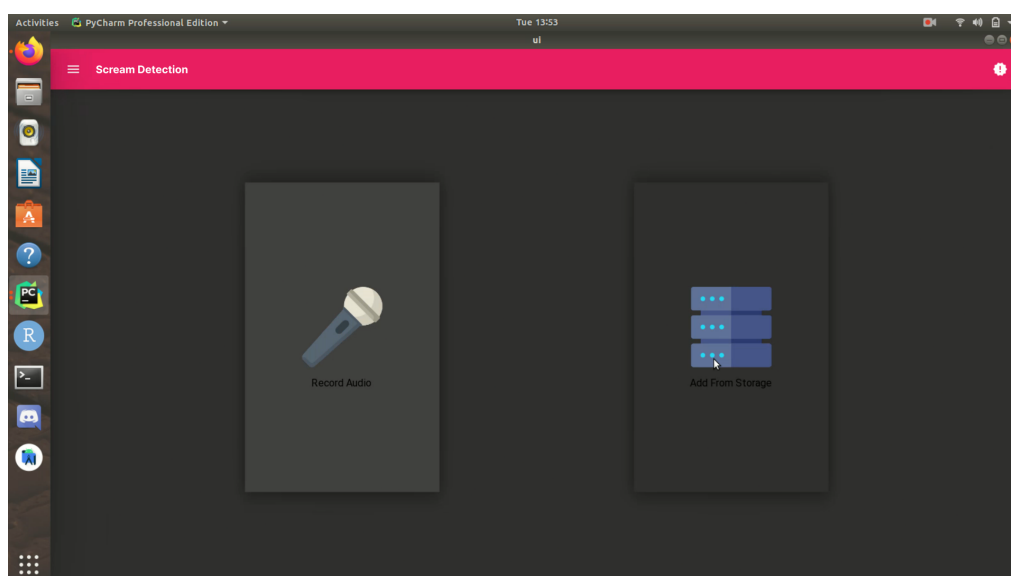


Figure 6.1: HOME Page

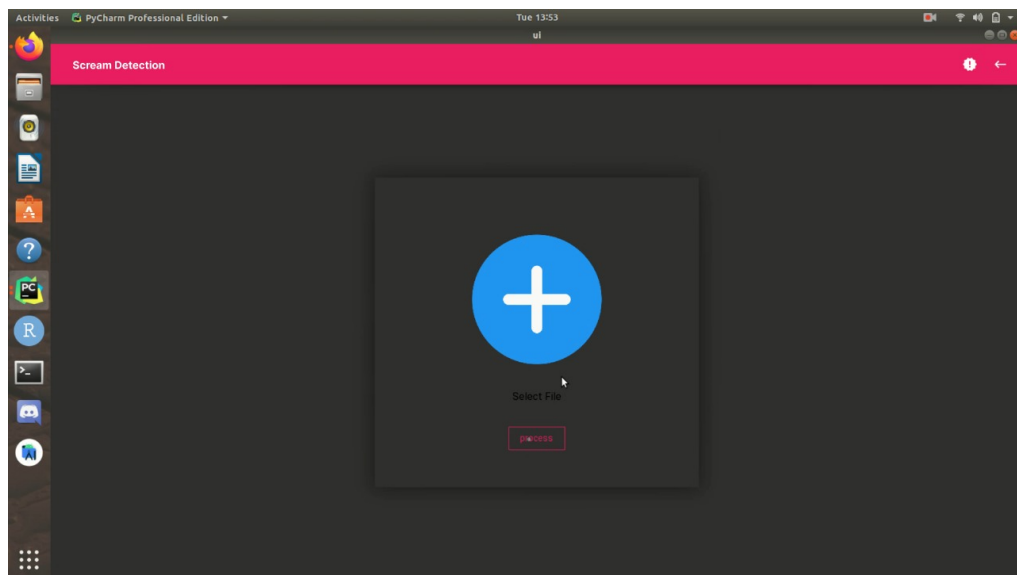


Figure 6.2: INPUT Page

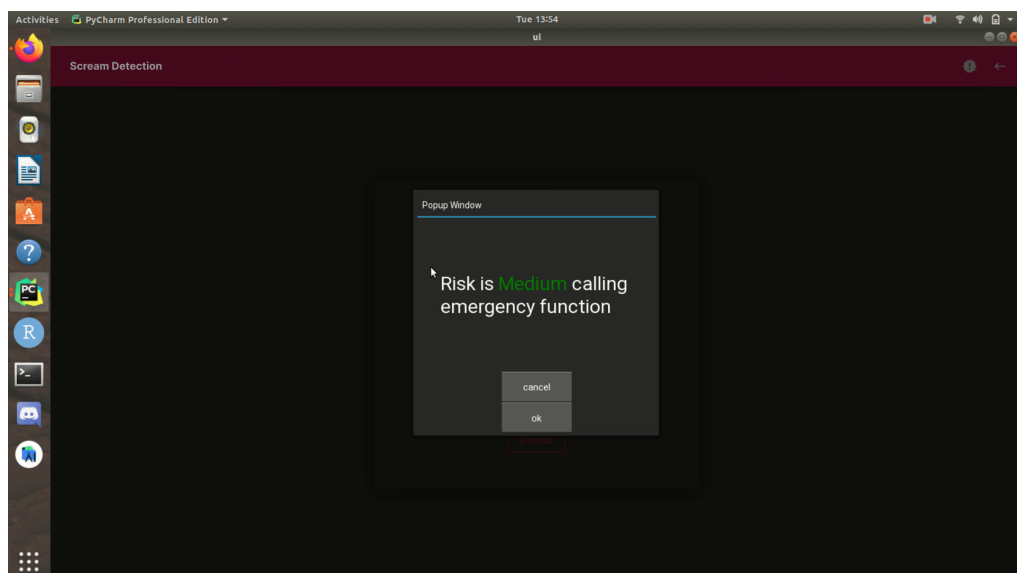


Figure 6.3: OUTPUT

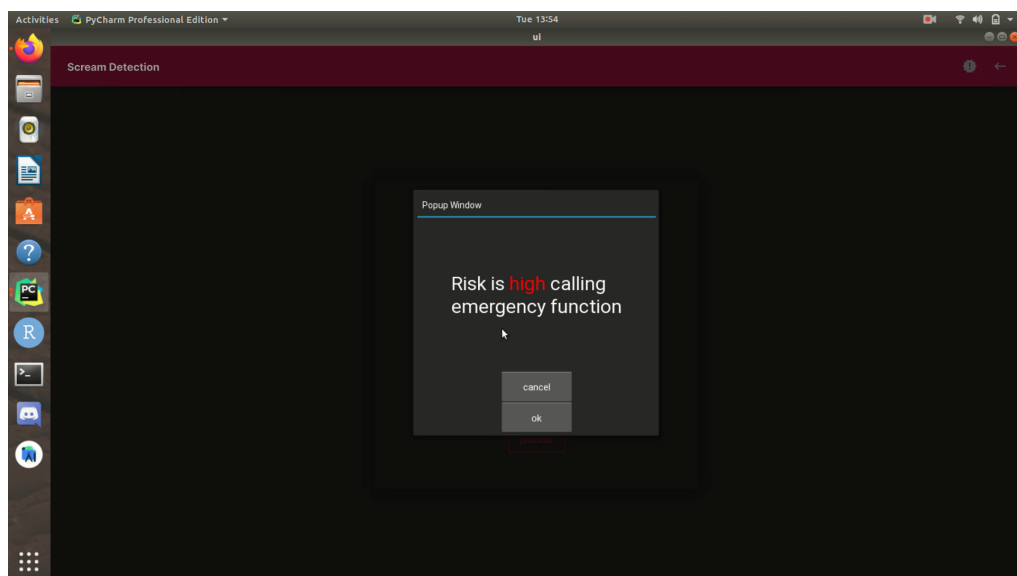


Figure 6.4: OUTPUT

Chapter 7

CONCLUSION AND SCOPE OF FUTURE WORK

7.1 Conclusion

This study demonstrates the successful integration of machine learning and deep learning methods for real-time human scream and distress noise detection. By combining SVM with MLP, the system achieves high accuracy and reliability in crime detection and emergency scenarios. The SVM model follows a three-step classification process: distinguishing human-generated sounds from background noise, recognizing distress signals, and differentiating screams from shouts. Using MFCCs as feature vectors ensures accurate classification by capturing the unique characteristics of distress sounds.

The MLP, built with six dense layers and activation functions like ReLU and sigmoid, enhances classification performance. It employs cross-entropy as the loss function and the Adam optimizer for weight adjustments, ensuring precise results. Standardizing audio data input size further improves compatibility and robustness. By integrating both models, the system reduces misclassifications, with SVM efficiently handling classification tasks and MLP managing complex patterns in large datasets.

Python libraries like Librosa for MFCC extraction and Keras for deep learning implementation simplify the development process. This hybrid approach enhances the reliability and scalability of distress sound detection, making it suitable for security, public safety, and emergency response applications. The study lays a strong foundation for future audio-driven crime detection systems, offering a practical and effective solution.

7.2 Future Scope

- **Enhanced Detection Capabilities:** Future iterations could incorporate more sophisticated audio processing algorithms to improve detection accuracy in complex environments, making the system more effective across diverse acoustic conditions through advancements in machine learning and signal processing.
- **Multi-Model System Integration:** Developing modules that combine audio detection with video surveillance and sensor networks can enhance overall security system effectiveness. This integration would enable more accurate threat detection and validation through multiple data sources.
- **Real-Time Location Tracking:** A precise location tracking module could provide immediate identification of sound sources within monitored areas, enabling faster emergency response times and more effective security interventions in large spaces like shopping malls or university campuses.
- **Adaptive AI Models:** Developing AI-driven detection models that automatically adapt to varying environmental conditions and noise levels could create a more robust system that maintains high accuracy across different scenarios and acoustic environments.
- **Pattern Analysis Enhancement:** Integration of historical data analysis with real-time detection could help identify patterns in security incidents, enabling predictive analytics for improved crime prevention and resource allocation strategies.
- **Mobile Application Integration:** Development of mobile applications for security personnel could enable instant alert notifications and location data access, improving response times and coordination during emergencies.

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APPENDIX

The Human Scream Detection System is an intelligent framework developed to classify audio signals into noise, speech, shout, or scream using ML and DL algorithms. The system leverages audio signal processing techniques and dual-stage classification models to detect high-risk auditory events. Designed with an intuitive Kivy-based user interface, it allows users to record or upload audio, process it, and receive emergency alerts based on the output classification.

Program Code Sections

- **Pgm code for MFCC Feature Extraction**
- **Pgm code for SVM Based Model**
- **Pgm code for MLP Based Model**
- **Pgm code for Main**
- **Pgm code for UI**

MFCC Feature Extraction

```
import pandas as pd
import numpy as np
import librosa

data = pd.read_csv('D:/Dev_stuff/metadata/UrbanSound8K.csv')
x_train = []
x_test = []
y_train = []
y_test = []

for i in range(len(data)):
    file = data.iloc[i]['slice_file_name']
    label = data.iloc[i]['classid']
    filename = "D:/Dev_stuff/audio_dataset/" + file
    y, sr = librosa.load(filename)
    mfccs = np.mean(librosa.feature.mfcc(y, sr, n_mfcc=40).T, axis=0)
    x_train.append(mfccs)
    y_train.append(label)
    if label == 1:
        sol = label

y, sr = librosa.load("D:/Dev_stuff/audio_dataset/damm_0.wav")
mfccs = np.mean(librosa.feature.mfcc(y, sr, n_mfcc=40).T, axis=0)
x_test.append(mfccs)
y_test.append(sol)

x_train = np.array(x_train)
x_test = np.array(x_test)
y_train = np.array(y_train)
y_test = np.array(y_test)

np.savetxt('train_data.csv', x_train, delimiter=',')
np.savetxt('test_data.csv', x_test, delimiter=',')
np.savetxt('train_labels.csv', y_train, delimiter=',')
np.savetxt('test_labels.csv', y_test, delimiter=',')
```

SVM Based Model

```
import librosa
import numpy as np
import pickle

def extract_features(filename):
```

```

y, sr = librosa.load(filename)
mfccs = np.mean(librosa.feature.mfcc(y, sr, n_mfcc=40).T, axis=0)
return np.array([mfccs])

def classify_audio(filename):
    phase1 = pickle.load(open('svm_based_model/phase1_model.sav', 'rb'))
    phase2 = pickle.load(open('svm_based_model/phase2_model.sav', 'rb'))
    phase3 = pickle.load(open('svm_based_model/phase3_model.sav', 'rb'))

    features = extract_features(filename)
    result1 = phase1.predict(features)

    if result1[0] == 2:
        result2 = phase2.predict(features)
        if result2[0] == 1:
            result3 = phase3.predict(features)
            return "Scream" if result3[0] == 1 else "Shout"
        else:
            return "Speech"
    else:
        return "Noise"

```

MLP Based Model

```

from keras.models import Sequential
from keras.layers import Dense
import pandas as pd

df = pd.read_csv('newresources.csv', index_col=0, engine='c')
X = df.iloc[:, 1:]
y = df.iloc[:, 0]

model = Sequential()
model.add(Dense(12, input_dim=X.shape[1], activation='relu'))
model.add(Dense(8, activation='relu'))
model.add(Dense(10, activation='relu'))
model.add(Dense(5, activation='relu'))
model.add(Dense(3, activation='relu'))
model.add(Dense(1, activation='sigmoid'))

model.compile(loss='binary_crossentropy', optimizer='adam', metrics=['accuracy'])
model.fit(X, y, validation_split=0.33, epochs=150, batch_size=50)

model.save('.')

```

Main

```

import sounddevice as sd
from scipy.io.wavfile import write
import threading
from kivymd.toast import toast
from kivy.uix.popup import Popup

def thread_for_rec(self):
    fs = 44100
    seconds = 10
    self.myrecording = sd.rec(int(seconds * fs), samplerate=fs,
    sd.wait()
    if not self.externally_stopped:
        write('recorded.wav', fs, self.myrecording)
        self.filename = "recorded.wav"

def mic_clicked(self):
    if self.recscreen.micbutton.source == "resources/icons/micof
        self.recscreen.micbutton.source = "resources/icons/micon
        self.externally_stopped = False
        threading.Thread(target=self.thread_for_rec()).start()
    else:
        sd.stop()
        self.externally_stopped = True
        write('recorded.wav', 44100, self.myrecording)
        self.filename = "recorded.wav"
        self.recscreen.micbutton.source = "resources/icons/micof

def process_the_sound(self):
    from modelloader import process_file
    from svm_based_model.model_loader_and_predict import svm_pro

    output1 = svm_process(self.filename)
    output2 = process_file(self.filename)

    if output1 == True and output2 == True:
        text = "[size=30]Risk is [color=#FF0000]high[/color] cal
        self.show_popup(text)
    elif output1 == True or output2 == True:
        text = "[size=30]Risk is [color=#008000]Medium[/color] c
        self.show_popup(text)
    else:
        toast("you are safe")

def show_popup(self, text):
    show = PopupWarning()
    show.label_of_emergency.text = text

```

```
self.popupWindow = Popup(title="Popup Window", content=show,
self.popupWindow.open())
```

UI

```
<AudioRecWindow>:
    micbutton: micbutton
    BoxLayout:
        orientation: 'vertical'

        MDToolbar:
            title: "Big Data App"
            elevation: 10
            right_action_items: ["keyboard-backspace", lambda x

        BoxLayout:
            orientation: 'vertical'
            MDCard:
                orientation: "vertical"
                BoxLayout:
                    ImageButton:
                        id: micbutton
                        source: "resources/icons/micoff.png"
                        on_release: app.mic_clicked()
                    BoxLayout:
                        MDRectangleFlatButton:
                            text: "Process"
                            on_release: app.process_the_sound()
                            pos_hint: {'center_x': .5}

<InternalStorageWindow>:
    BoxLayout:
        orientation: 'vertical'

        MDToolbar:
            title: "Big Data App"
            elevation: 10
            right_action_items: ["keyboard-backspace", lambda x

        BoxLayout:
            MDCard:
                orientation: "vertical"
                BoxLayout:
                    ImageButton:
                        source: "resources/icons/plus.png"
                        on_release: app.internalstoragescreen_to
```

```
BoxLayout:
    MDLabel:
        text: "Select File"
        halign: 'center'
    MDRectangleFlatButton:
        text: "Process"
        on_release: app.process_the_sound()
        pos_hint: {'center_x': .5}

<PopupWarning>:
    label_of_emergency: label_of_emergency
    BoxLayout:
        orientation: "vertical"

    Label:
        id: label_of_emergency
        text: ""
        markup: True
        size_hint: 0.6, 0.2
        pos_hint: {"center_x": .5}

    Button:
        text: "Cancel"
        size_hint: 0.3, 0.04
        on_press: app.close_popup()
        pos_hint: {"center_x": .5}

    Button:
        text: "Ok"
        size_hint: 0.3, 0.04
        on_press: app.close_popup()
        pos_hint: {"center_x": .5}
```



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